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The Affine Symmetric Group

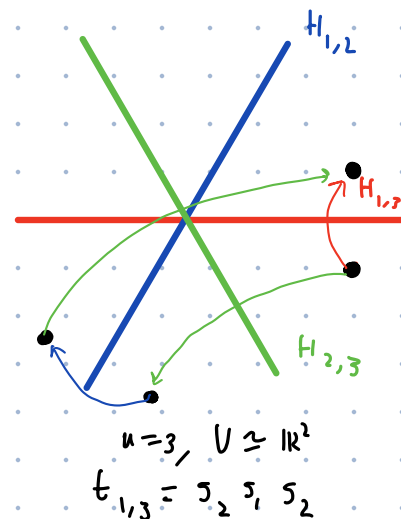
Defn: $V := \{x \in \mathbb{R}^n \mid \sum_{i=1}^n x_i = 0\}$

For $1 \leq i < j \leq n$, consider $H_{i,j} = \{x \in V \mid x_i - x_j = 0\}$
 $t_{i,j}$ = reflection fixing $H_{i,j}$

$$\tilde{G}_n = \langle t_{i,j} \mid 1 \leq i < j \leq n \rangle = \langle t_{i,i+1} \mid 1 \leq i \leq n-1 \rangle$$

$\tilde{G}_n$ is also a Coxeter group

$$= \left\langle s_1, \dots, s_{n-1} \mid \begin{array}{l} s_i^2 = e, \quad s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \\ s_i s_j = s_j s_i \text{ if } |i-j| \geq 2 \end{array} \right\rangle$$



We now define the Affine Symmetric Group analogously:

For $1 \leq i < j \leq n, p \in \mathbb{Z}$ consider $H_{i,j}^p = \{x \in V \mid x_i - x_j = p\}$
 $t_{i,j}^p$ = reflection fixing $H_{i,j}^p$

Defn: $\tilde{\tilde{G}}_n := \langle t_{i,j}^p \mid 1 \leq i, j \leq n, p \in \mathbb{Z} \rangle$

Defn: An alcove is a connected component of $V \setminus \bigcup_{i,j,p} H_{i,j}^p$

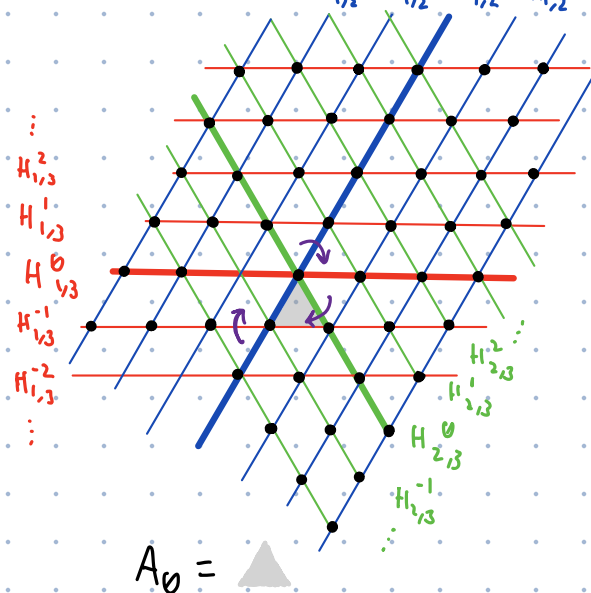
Fix one to be A_0

$\tilde{G}_n$ acts on the set of alcoves by permuting them

transitively & faithfully
 $\Rightarrow$ set of alcoves $\xleftrightarrow{\sim} \tilde{\tilde{G}}_n$

$$\tilde{\tilde{G}}_n = \langle t_{(1,n)}^{-1}, t_{i,i+1}^{s_i} \mid 1 \leq i \leq n-1 \rangle$$

These gens are marked w/ $\curvearrowright$

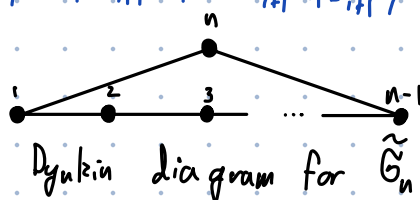


The root lattice of V is the set of integer points. (intersection points • of lines)

One can say $\tilde{\tilde{G}}_n$ is the grp of rigid transformations preserving the root lattice.

Prop: $\tilde{G}_n$ is a Coxeter group:

$$\tilde{G}_n = \langle s_1, \dots, s_n \mid s_i^2 = 1, \overbrace{s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}}^{\text{indexing modulo } n}, s_i s_j = s_j s_i \text{ for } |i-j| \geq 2 \rangle$$



and a group of permutations:

$$\tilde{G}_n = \left\{ \text{bijections } u: \mathbb{Z} \rightarrow \mathbb{Z} \mid u(i+n) = u(i) + n, \sum_{i=1}^n u_i = \binom{n+1}{2} \right\}$$

$$\tilde{G}_n = \text{Stab}([n]) = \{ u \in \tilde{G}_n \mid \{u(1), \dots, u(n)\} = [n] \}$$

one can view $\tilde{G}_n$ as $\text{Sym}(\mathbb{Z}) / \{s_i \sim s_{i+n} \mid i \in \mathbb{Z}\}$

We can write elements of $\tilde{G}_n$ as $u = [a_1, \dots, a_n]$, $u(i) = a_i$
 $s_i = [1, 2, \dots, i+1, i, \dots, n]$ $1 \leq i \leq n-1$, $s_n = [0, 2, \dots, n-1, n+1]$
 With this notation

$$\begin{aligned} [a_1, \dots, a_n] s_i &= [a_1, \dots, a_{i+1}, a_i, \dots, a_n] \\ [a_1, \dots, a_n] s_n &= [a_n - n, a_2, \dots, a_{n-1}, a_1 + n] \end{aligned}$$

Defn: We may write any $u \in \tilde{G}_n$ as $s_{i_1} \dots s_{i_\ell}$.
 Let $\ell(u)$ be the minimum such ℓ .
 Same defn as G_n , or any Coxeter group

Prop: $u \in \tilde{G}_n$, $u = [a_1, \dots, a_n]$

$$\begin{aligned} \ell(u) &= \# \{ (i, j) \in [n] \times [n] \mid i < j, a_i > a_j \} + \sum_{1 \leq i < j \leq n} \left\lfloor \frac{|a_j - a_i|}{n} \right\rfloor \\ &= \# \{ (i, j) \in [n] \times \mathbb{Z}_{>0} \mid i < j, a_i > a_j \} \end{aligned}$$

For $u \in G_n$, $\ell(u) = \# \{ (i, j) \in [n] \times [n] \mid i < j, a_i > a_j \}$

Route: With the bijection $\tilde{G}_n \leftrightarrow$ alcoves, the length $\ell(u)$ may be computed as the number of hyperplanes $H_{i,j}$ crossed along a shortest path from the alcove corresponding to e to the alcove corresponding to u .

Recall: $T = \{t_{i,j} \mid 1 \leq i < j \leq n\}$ the transpositions of $\mathfrak{S}_n$

For $u, v \in \mathfrak{S}_n$, say $u \xrightarrow{(t)} v$ if $\exists t \in T, v = ut$, and $\ell(v) = \ell(u) + 1$
 Equiv, $\exists 1 \leq i < j \leq n, u(i) < u(j), v = ut_{i,j}$

Burkhat order is given by $u \leq v$ if $\exists u_0, \dots, u_k, u = u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_k = v$

Prop: Burkhat order can be computed via the diagram of $u \in \mathfrak{S}_n$

Defn: for $i, j \in [n]$
 $r_u(i, j) := \# \{k \leq i \mid w(k) \leq j\}$ is the rank function for

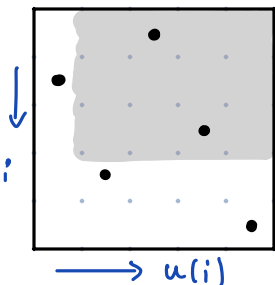


diagram for $u = 31425$
 $r_u(3, 2) = 2$

Then $u \leq v \Leftrightarrow \forall 1 \leq i, j \leq n, r_u(i, j) \leq r_v(i, j)$

Defn: For $i \not\equiv j \pmod n$, say $t_{i,j} := \prod_{k \in \mathbb{Z}} (i + kn, j + kn)$
 $\neq [a_1=1, \dots, a_i=j, \dots, a_j=i, \dots, a_n=n]$

$\tilde{T} := \{t_{i, j+kn} \mid 1 \leq i < j \leq n, k \in \mathbb{Z}\}$

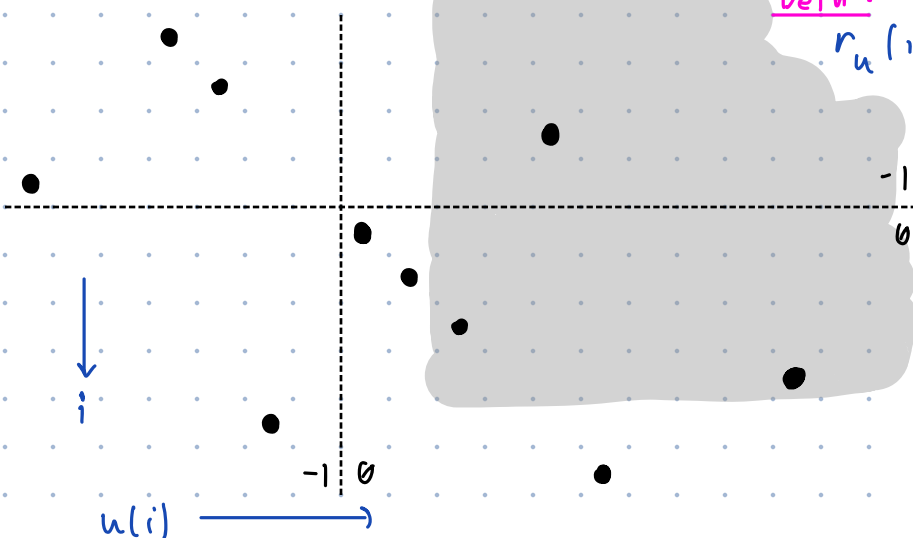
For $u, v \in \tilde{\mathfrak{S}}_n$, say $u \xrightarrow{(t)} v$ if $\exists t \in \tilde{T}, v = ut$, and $\ell(v) = \ell(u) + 1$
 Equiv, $\exists i, j \in \mathbb{Z}, i < j, i \not\equiv j \pmod n, u(i) < u(j), v = ut_{i,j}$

Burkhat order is given by $u \leq v$ if $\exists u_0, \dots, u_k, u = u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_k = v$

Prop: Burkhat order can be computed via the diagram of $u \in \tilde{\mathfrak{S}}_n$

Defn: for $i, j \in \mathbb{Z}$
 $r_u(i, j) := \# \{k \leq i \mid w(k) \leq j\}$

Then $u \leq v$
 $\Leftrightarrow \forall i, j \in \mathbb{Z}, r_u(i, j) \leq r_v(i, j)$

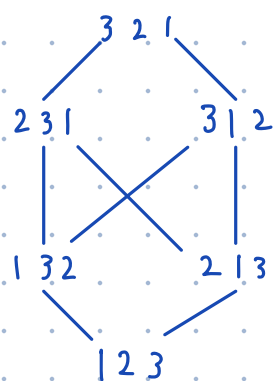


$u = [1 \ 2 \ 9 \ -2 \ 5]$

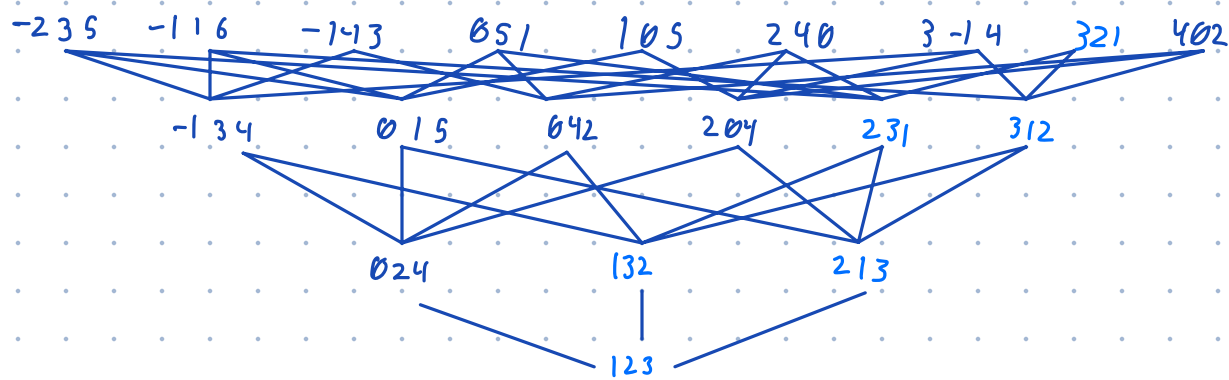
$r_u(3, 2) = 3$

Rank: $G_n \subseteq \tilde{G}_n$, and $u \leq v$ in $G_n \Leftrightarrow u \leq v$ in $\tilde{G}_n$

Ex: We can compute the Hasse diagrams for the Bruhat order



Bruhat order for G_3



Bruhat for $\tilde{G}_3$ for elements of rank ≤ 3
 G_3 highlighted

< We'll see how much time there is left >

Defn: Parabolic Subgroups are subgroups of Coxeter groups generated by subsets of the Coxeter generators

Defn: $G_n = \langle s_1, \dots, s_{n-1} \rangle = \langle S \rangle$

$$(G_n)_k = \langle s_1, \dots, s_{k-1}, s_{k+1}, \dots, s_{n-1} \rangle \quad \text{maximal parabolic}$$

$$= \text{stab}([k]) \cong G_k \times G_{n-k}$$

Let $G_n^{(k)} := \{ w \in G_n \mid w(1) < \dots < w(k), w(k+1) < \dots < w(n) \}$ unique descent at k

Then every element $w \in G_n$ has a unique decomposition

$$w = u \cdot v,$$

$$u \in G_n^{(k)}, v \in (G_n)_k$$

$$l(w) = l(u) + l(v)$$

So $G_n^{(k)}$ gives coset reps of $G_n / (G_n)_k$

Furthermore, $G_n^{(k)}$ is the set of coset reps of minimal length.

The story for $\tilde{G}_n$ is the same.

Defn: $\tilde{G}_n = \langle s_1, \dots, s_n \rangle$

$$(\tilde{G}_n)_k^{\wedge} = \langle s_1, \dots, s_{k-1}, s_{k+1}, \dots, s_n \rangle = \text{stab}([k+1, \dots, n+k])$$

$$(\tilde{G}_n)_n^{\wedge} = G_n$$

$$\tilde{G}_n^{(k)} = \{ w \in \tilde{G}_n \mid w(1) < \dots < w(k), w(k+1) < \dots < w(n+1) \}$$

We enjoy the same decomposition and coset representation.

Ex: We consider $\tilde{G}_n / G_n$
To find the coset rep of $w \in \tilde{G}_n$, we want the element of minimal length. This unique element u such that

$$l(u s_i) < l(u) \iff i = n$$

i.e. in window notation, $u = [u_1, \dots, u_n]$ satisfies $u_1 < \dots < u_n$

As another more concrete example, consider $n=5$ and $\tilde{G}_5 / (\tilde{G}_5)_3^{\wedge}$
We compute a coset representative of $w(\tilde{G}_5)_3^{\wedge}$

$$\text{Let } w = [-3, 6, 3, -5, 14]$$

The rule is to write the entries $\{w(ln+k+1), \dots, w(ln+k+n)\}$ in inc. order

$$l=0 \quad \left\{ \begin{array}{ccccc} w(4) & w(5) & w(6) & w(7) & w(8) \\ -5 & 14 & 2 & 11 & 8 \end{array} \right\}$$

$$\begin{array}{ccccc} -5 & 2 & 8 & 11 & 14 \\ \hline 3 & 6 & 9 & -5 & 2 \end{array} \quad \begin{array}{l} \text{rearrange} \\ \text{subtract 5 to get } w(1), \dots, w(5) \end{array}$$

Sources:

[AL 01] Alain Lascoux - Ordering the Affine Symmetric Group

[BB 05] Anders Björner, Francesco Brenti - Combinatorics of Coxeter Groups

[Yas 24] Jad Abou Yassin - Affine Symmetric Groups